

by-Vaibhav Sahu

AI Prompt Guide

for Functional & Automation Test Engineers

Your Daily AI Assistant for Testing Excellence

Version 1.0 | January 2025

by-Vaibhav Sahu

Table of Contents

by-Vaibhav Sahu

Introduction

This comprehensive guide helps functional and automation test engineers leverage AI assistance effectively in their daily work. Whether you're writing test cases, creating automation scripts, debugging issues, or generating documentation, this guide provides ready-to-use prompt templates for every common testing scenario.

Each section includes practical prompt templates that you can copy, customize, and use immediately to boost your productivity and testing effectiveness.

1. Test Planning & Strategy

Creating Test Plans

Prompt Template:

```
I need to create a test plan for [feature/application name].Context:-  
Application type: [web/mobile/API/desktop]- Technology stack:  
[Java/Python/.NET, etc.]- Testing scope: [functional/integration/E2E]-  
Timeline: [duration]- Team size: [number]Please provide:1. Test strategy  
approach2. Test scope and out of scope items3. Risk assessment4. Resource  
allocation5. Timeline with milestones
```

Test Coverage Analysis

Prompt Template:

```
Analyze test coverage for [feature/module name].Current coverage:- [List  
existing test scenarios]Requirements:- [Paste requirements or user  
stories]Identify:1. Missing test scenarios2. Edge cases not covered3. Negative  
test scenarios needed4. Priority gaps (high/medium/low)
```

2. Test Case Design

Writing Test Cases from Requirements

Prompt Template:

```
Convert the following requirement into detailed test cases:Requirement:[Paste requirement/user story]Format needed: [Gherkin/Traditional/BDD]Include:- Test case ID convention- Preconditions- Test steps- Expected results- Priority and severity- Positive and negative scenarios
```

Converting Manual Tests to Automation

Prompt Template:

```
I have the following manual test case that needs automation:[Paste manual test case]Convert this to:- Framework: [Selenium/Cypress/Playwright/RestAssured]- Language: [Java/Python/JavaScript]- Pattern: [Page Object Model/Screenplay]Include:1. Code structure2. Reusable methods3. Assertions4. Error handling5. Comments explaining the logic
```

3. Test Automation

Selenium WebDriver (Java)

Prompt Template:

Create a Selenium WebDriver script in Java for: Scenario: [Describe test scenario] Framework: [TestNG/JUnit] Pattern: Page Object Model Requirements:- Browser: [Chrome/Firefox/Edge]- Waits: Explicit waits with expected conditions- Assertions: [TestNG/JUnit assertions]- Include comments explaining each section- Handle common exceptions- Add logging statements Provide: 1. Page Object class 2. Test class 3. Base test setup

API Automation (RestAssured)

Prompt Template:

Create RestAssured API test for: Endpoint: [HTTP method] [URL] Request:- Headers: [list headers]- Body: [JSON/XML payload] Validations needed:- Status code- Response time- Response body fields- [Other validations] Include:- Request specification- Response specification- JSON Schema validation- Comments for each validation

Framework Setup

Prompt Template:

Help me set up a [Selenium/Cypress/Playwright/RestAssured] automation framework. Requirements:- Language: [Java/Python/JavaScript/TypeScript]- Build tool: [Maven/Gradle/npm]- Test runner: [TestNG/JUnit/Mocha]- Reporting: [Extent Reports/Allure/Mochawesome]- CI/CD: [Jenkins/GitLab/GitHub Actions] Provide: 1. Project structure 2. Configuration files 3. Base classes/utilities 4. Example test 5. README with setup instructions

4. Defect Management

Writing Effective Bug Reports

Prompt Template:

Help me write a detailed bug report for: Issue: [Brief description] Steps to reproduce: [List steps] Expected result: [What should happen] Actual result: [What actually happened] Environment: [Browser/OS/Version] Create a bug report with:- Clear title- Severity and priority assessment- Detailed steps- Impact analysis- Suggested workaround if any

Bug Analysis

Prompt Template:

Analyze this bug pattern: Bugs found: [List of related bugs] Provide: 1. Root cause analysis 2. Common patterns 3. Areas needing attention 4. Preventive measures 5. Test coverage gaps

5. Test Data Management

Generating Test Data

Prompt Template:

```
Generate test data for [scenario/entity].Entity: [User/Transaction/Product, etc.]Fields needed:- [Field 1: type and constraints]- [Field 2: type and constraints]Requirements:- Format: [JSON/CSV/SQL/Excel]- Number of records: [count]- Include edge cases: [Yes/No]- Valid/Invalid mix: [ratio]Include data for:- Positive scenarios- Boundary values- Negative scenarios
```

SQL Queries for Data Validation

Prompt Template:

```
Create SQL queries for data validation:Scenario: [What needs to be validated]Tables: [List relevant tables and their relationships]Validation criteria: [What to check]Provide:1. Data extraction queries2. Validation queries3. Cleanup queries4. Comments explaining each query
```

6. API Testing

API Test Scenarios

Prompt Template:

```
Create comprehensive API test scenarios for:API Endpoint: [METHOD]
[URL]Purpose: [What the API does]Provide test cases for:1. Positive scenarios2.
Negative scenarios3. Boundary conditions4. Security testing (auth, injection)5.
Performance considerations6. Status code validations7. Response schema
validation
```

Security Testing Scenarios

Prompt Template:

```
Create security test scenarios for:Application: [Type and
technology]Authentication: [OAuth/JWT/Basic/API Key]Include tests for:-
Authentication bypass- Authorization checks- Input validation- SQL injection-
XSS- CSRF- Rate limiting- Sensitive data exposure
```

7. Performance Testing

JMeter/K6 Script Creation

Prompt Template:

```
Create a [JMeter/K6] performance test script for:Scenario: [Load test scenario]Target: [URL/Endpoint]Load profile:- Users: [concurrent users]- Ramp-up: [time]- Duration: [test duration]Include:- HTTP samplers/requests- Think time- Assertions- Listeners/output format- Comments explaining load pattern
```

Performance Test Analysis

Prompt Template:

```
Analyze these performance test results:Metrics:- Response time: [values]- Throughput: [requests/sec]- Error rate: [percentage]- Resource utilization: [CPU/Memory]Provide:1. Performance bottleneck analysis2. Comparison with benchmarks3. Recommendations4. Areas for optimization
```

8. Code Review & Optimization

Code Review

Prompt Template:

```
Review this automation code:[Paste code]Framework: [Framework name]Language: [Language]Check for:1. Best practices adherence2. Code readability3. Reusability4. Performance issues5. Hard-coded values6. Exception handling7. Wait strategies8. Assertion effectiveness9. Comment qualityProvide specific improvement suggestions with examples.
```

Code Optimization

Prompt Template:

```
Optimize this test automation code:[Paste code]Focus on:- Execution speed-Maintainability- Reusability- Reduced flakiness- Better error handlingProvide:1. Optimized version with comments2. Explanation of changes3. Performance impact
```

9. Documentation

Test Strategy Document

Prompt Template:

```
Create a test strategy document for:Project: [Project name]Type: [Web/Mobile/API/Desktop]Duration: [Timeline]Include:1. Testing scope and objectives2. Test approach (types of testing)3. Entry/exit criteria4. Tools and frameworks5. Roles and responsibilities6. Risk and mitigation7. Test environment requirements8. Deliverables and timeline
```

README for Automation Framework

Prompt Template:

```
Create a comprehensive README for our automation framework.Framework details:-Technology: [Stack details]- Structure: [Folder structure]- Tools: [List of tools]Include:1. Project overview2. Prerequisites3. Installation steps4. Configuration guide5. Running tests (local and CI/CD)6. Report generation7. Troubleshooting common issues8. Contributing guidelines9. Code examples
```

10. Learning & Troubleshooting

Learning New Tools/Frameworks

Prompt Template:

I need to learn [Tool/Framework name].My background:- Current expertise: [Your experience]- Programming language: [Known languages]- Time available: [Learning timeline]Provide:1. Learning roadmap2. Key concepts to master3. Practical exercises4. Best practices5. Common pitfalls to avoid6. Useful resources

Debugging & Troubleshooting

Prompt Template:

Help me debug this issue:Problem: [Describe the issue]Code/Script: [Paste relevant code]Error message: [Paste error]Environment: [Browser/OS/Tool versions]What I've tried: [List attempted solutions]Provide:1. Root cause analysis2. Step-by-step solution3. Preventive measures4. Best practices to avoid similar issues

Pro Tips for Effective Prompts

Do's

- **Be Specific:** Include framework, language, and context
- **Provide Examples:** Show current code or approach
- **State Constraints:** Mention limitations or requirements
- **Ask for Explanations:** Request comments in code
- **Iterate:** Refine prompts based on responses
- **Include Error Messages:** Share complete error logs
- **Specify Format:** Define desired output format

Don'ts

- **Avoid Vague Requests:** "Make it better" without context
- **Don't Skip Context:** Framework and environment matter
- **Don't Assume Knowledge:** Specify your expertise level
- **Don't Ignore Security:** Sanitize sensitive data before sharing
- **Don't Accept Blindly:** Understand and validate suggestions

by-Vaibhav Sahu

Quick Reference - Common Daily Prompts

Task	Quick Prompt
Morning Standup Prep	Summarize my testing progress for standup:- Completed: [List tasks]- In progress: [Current work]- Blockers: [Issues]Format as brief talking points.
Quick Bug Report	Convert my notes to a bug report:[Paste rough notes]Make it professional and comprehensive.
Fast Test Case Generation	Generate 10 test cases for: [Feature]Priority: High priority scenarios onlyFormat: Gherkin
Code Snippet	Quick [Selenium/RestAssured] code for: [Action]Language: [Language]Just the essential code with comments, no explanation.
Regex Helper	Create regex to validate: [Pattern description]Provide pattern and test cases.
XPath/CSS Selector	Best selector for element with: [Attributes]Provide XPath and CSS, explain which is better.

by-Vaibhav Sahu

Sample Daily Workflow with AI Prompts

9:00 AM - Planning

Review my test plan for [feature] and suggest:1. Missing test scenarios2. Risk areas3. Time estimation validation

10:00 AM - Automation

Create Page Object for [page name]:- Locators for [list elements]- Methods for [list actions]- Java with Selenium- Include comments and JavaDoc

2:00 PM - Debugging

This test is flaky: [paste code]Help identify:1. Possible race conditions2. Wait improvements needed3. Better selector strategies

4:00 PM - Review & Documentation

Review this automation code for:1. Code quality issues2. Best practices violations3. Improvement suggestionsThen help me document the changes in commit message format.

5:00 PM - Reporting

Create a test summary report from:- Total: [number] test cases- Passed: [number]- Failed: [number]- Blocked: [number]- Key findings: [list]Format: Executive summary for stakeholders

Conclusion

This prompt guide is your companion for leveraging AI assistance effectively in daily testing activities. Remember:

1. **Iterate:** Start broad, refine based on responses
2. **Learn:** Ask for explanations to build knowledge
3. **Validate:** Always review and test AI-generated code
4. **Customize:** Adapt prompts to your specific context
5. **Share:** Help your team use these templates

Happy Testing! 🚀

Version 1.0 | Last Updated: January 2025

This is a living document. Adapt and improve based on your experience!